# Zoho Analytics Integration

## 1. Overview

### 1.1 Description

**Zoho Analytics Integration** connects **Zoho Analytics** (business intelligence and reporting platform) with **Newo Platform**.

It enables:

- Verifying caller identity against a vendor database
- Looking up invoice status by load number (due date, payment status, amount due, payee)
- Verifying rate details by load number (rate, payee, payment amount)
- Identifying caller's company/location with fuzzy name matching

This integration is designed for **logistics companies** who need **an AI phone agent to answer vendor (carrier/transporter) inquiries about invoices and rates** using data stored in Zoho Analytics tables.

### 1.2 Use Cases

- When a vendor calls and asks about invoice status → Agent verifies identity, queries the invoices table, and reads back due date, payment status, amount due, and payee
- When a vendor calls to verify a rate → Agent queries the rates table and reads back rate, payee, and payment amount
- When a caller needs to be verified → Agent checks the vendors table by company name (and optionally phone number)
- When a caller mentions a company → Agent identifies the correct location from the locations table, handling multiple matches with disambiguation

### 1.3 Supported Features

| Feature | Supported | Notes |
|---|---|---|
| Invoice Status Lookup | ✅ | By load number + vendor name |
| Rate Verification | ✅ | By load number |
| Caller Identity Verification | ✅ | By vendor name; phone number optional |
| Location Identification | ✅ | LIKE matching for partial company names |
| Multi-Region Support | ✅ | US, EU, IN, AU, JP, SA, CA data centers |
| OAuth2 Authentication | ✅ | Standard redirect flow with token auto-refresh |
| Custom Canvas Scenarios | ✅ | Logistics-specific scenarios and intent types |
| Booking/Scheduling | ❌ | Not applicable — this is a data query integration |
| Write Operations | ❌ | Read-only — agent never modifies Zoho Analytics data |

## 2. Requirements

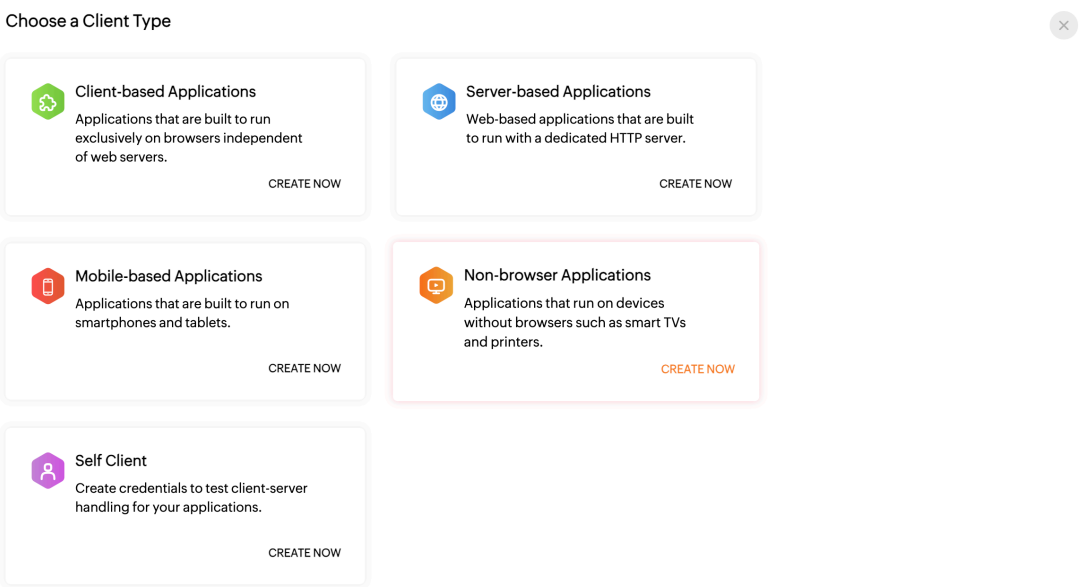
Before installation:

- Active **Zoho Analytics** account with API access (free tier works)
- **Server-based Applications** client registered in Zoho API Console
- A Zoho Analytics workspace with 4 tables: **Invoices**, **Rates**, **Vendors**, **Locations**
- Data synced from your TMS (Transportation Management System) to these tables

## 3. How to Setup

### 3.1 Step 1 — Register OAuth Client in Zoho

1. Go to Zoho API Console
2. Click **Add Client** → select **Server-based Applications**
3. Fill in:
   - **Client Name**: `Newo Zoho Analytics`
   - **Homepage URL**: `https://newo.ai`
   - **Authorized Redirect URIs**: `https://static.newo.ai/oauth/index.html`
4. Copy **Client ID** and **Client Secret**




### 3.2 Step 2 — Prepare Tables in Zoho Analytics

Create a workspace with 4 tables:

**Invoices** — invoice/payment data:

| Column Name | Data Type |
|---|---|
| Load Number | Plain Text |
| Vendor Name | Plain Text |

| | |
|---|---|
| Due Date | Date |
| Payment Status | Plain Text |
| Amount Due | Number |
| Payee | Plain Text |

**Rates** — rate and payment data:

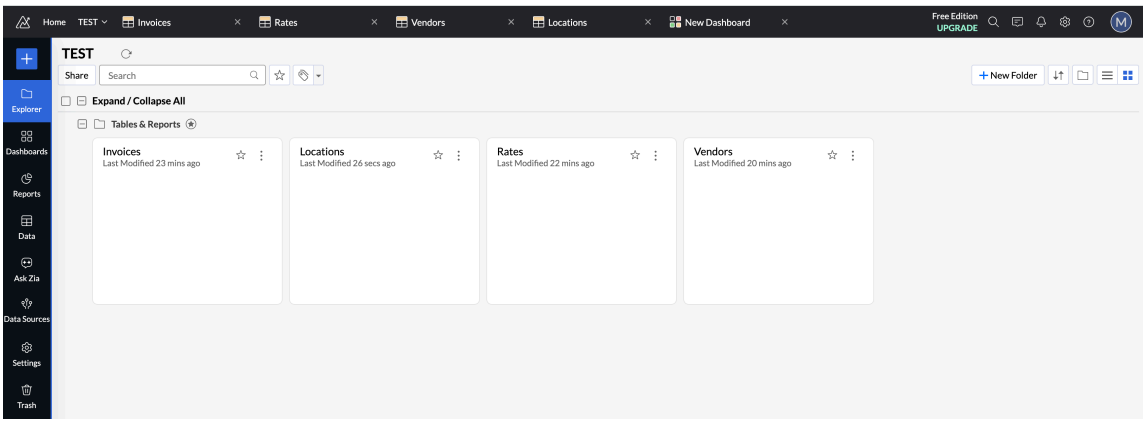
| Column Name | Data Type |
|---|---|
| Load Number | Plain Text |
| Rate | Number |
| Payee | Plain Text |
| Payment Amount | Number |

**Vendors** — vendor directory:

| Column Name | Data Type |
|---|---|
| Vendor Name | Plain Text |
| Phone | Plain Text |
| Contact Person | Plain Text |
| Email | Plain Text |

**Locations** — company/location directory:

| Column Name | Data Type |
|---|---|
| Company Name | Plain Text |
| Location | Plain Text |
| Region | Plain Text |
| Account ID | Plain Text |




> ***Important**: Column names must match exactly (case-sensitive, with spaces). The integration queries by these exact column names.*

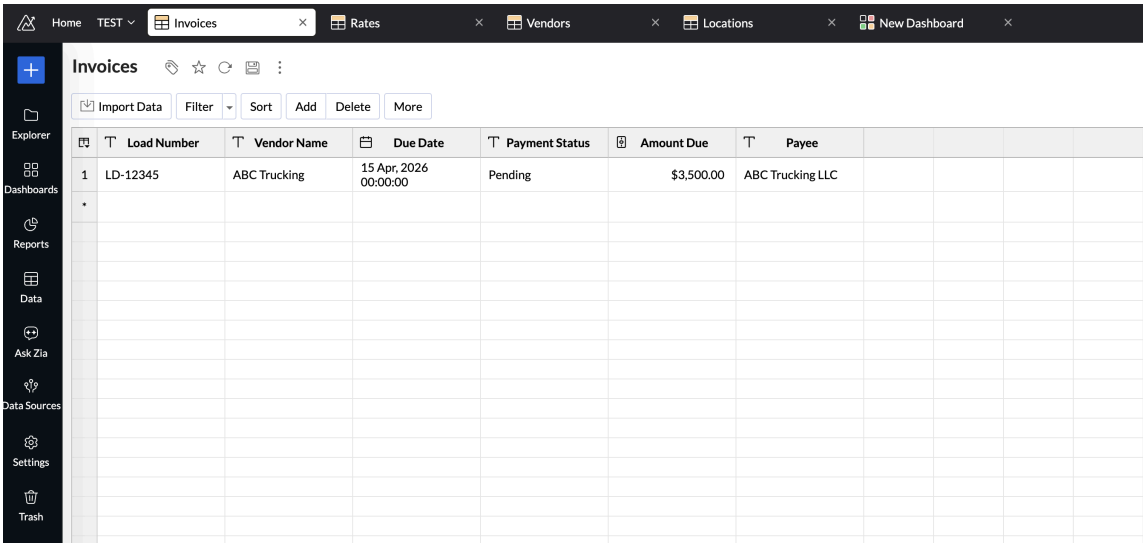
### 3.3 Step 3 — Get Workspace ID and View IDs

Open each table in Zoho Analytics. The View ID is visible in the browser URL:

```
https://analytics.zoho.eu/workspace/WORKSPACE_ID/view/VIEW_ID
```

Note down:

- **Workspace ID**
- **View ID** for each of the 4 tables



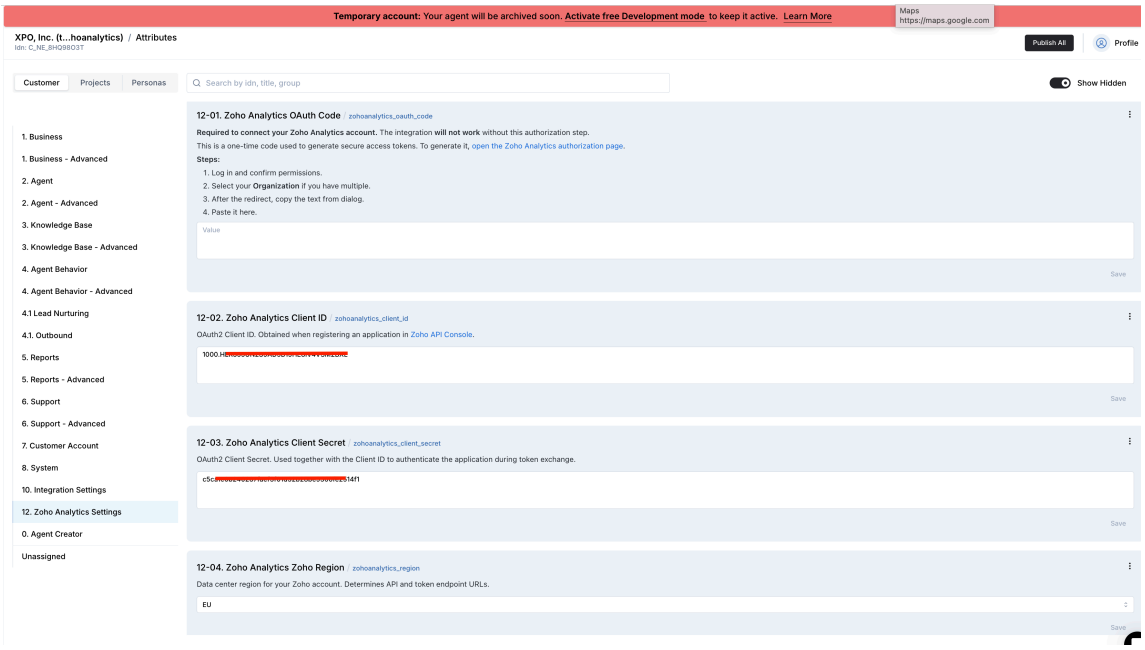

### 3.4 Step 4 — Get Organization ID

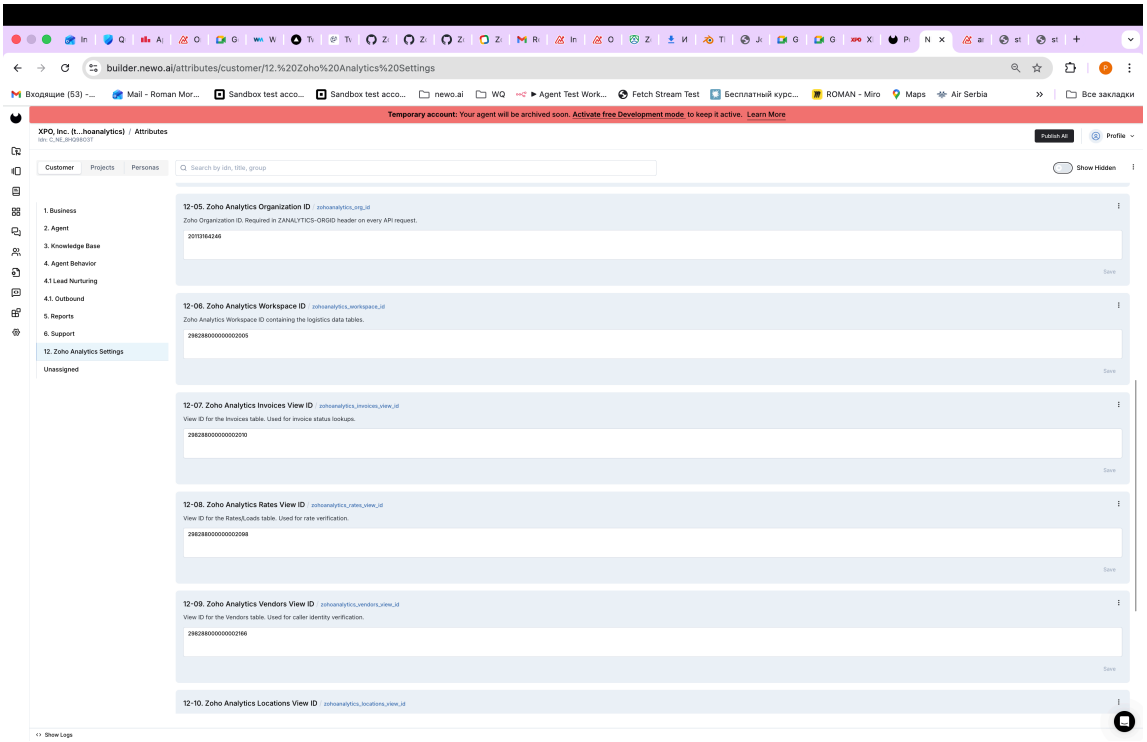
Go to **Zoho Analytics → Setup → General → Organization Details**. Copy the **Organization ID** from the URL or the settings page.

### 3.5 Step 5 — Connect in Newo Platform

1. Open **builder.newo.ai → Project → Attributes**
2. Set the following attributes:

| Attribute | Required | Description |
|---|---|---|
| `zohoanalytics_client_id` | ✅ | OAuth2 Client ID from Zoho API Console |
| `zohoanalytics_client_secret` | ✅ | OAuth2 Client Secret from Zoho API Console |
| `zohoanalytics_region` | ✅ | Data center region: US, EU, IN, AU, JP, SA, or CA |
| `zohoanalytics_org_id` | ✅ | Zoho Organization ID |
| `zohoanalytics_workspace_id` | ✅ | Zoho Analytics Workspace ID |
| `zohoanalytics_invoices_view_id` | ✅ | View ID for the Invoices table |
| `zohoanalytics_rates_view_id` | ✅ | View ID for the Rates table |
| `zohoanalytics_vendors_view_id` | ✅ | View ID for the Vendors table |
| `zohoanalytics_locations_view_id` | ❌ | View ID for the Locations table |
| `zohoanalytics_setup_scenarios` | ❌ | Auto-add logistics scenarios to canvas (default: True) |
| `zohoanalytics_override_agent_attributes` | ❌ | Override SA attributes on publish (default: True) |




3. **Fill in Client ID and Client Secret first**, then **Publish** the project
4. After publish, click the OAuth link in the **OAuth Code** attribute to authorize
5. Paste the code and **Publish** again

On publish, the SetupFlow automatically:

- Resolves regional API and token endpoints based on selected region
- Creates HTTP connectors for API requests and token refresh
- Exchanges OAuth code for access + refresh tokens

- Registers custom tools in ConvoAgent (verify caller, invoice lookup, rate verification, location identification)
- Adds logistics-specific scenarios and intent types to the Workflow Builder Canvas

# 4. How to Use

## 4.1 How the Integration Works

The integration operates based on **Triggers and Actions** via the event-driven system.

- A **Trigger** is a code-phrase the agent says, detected by the ConvoAgent tool caller
- An **Action** is a query to Zoho Analytics Sync Export API

**Available Triggers:**

| Trigger | Tool Event | Description |
|---|---|---|
| Verify Caller | `zohoanalytics_verify_caller_tool` | Agent says "Let me verify your information..." |
| Invoice Status Lookup | `zohoanalytics_query_invoice_status_tool` | Agent says "Let me look up the invoice..." |
| Rate Verification | `zohoanalytics_query_rate_verification_tool` | Agent says "Let me check the rate..." |
| Identify Location | `zohoanalytics_identify_location_tool` | Agent says "Let me look up that company..." |

**Available Actions:**

- **Verify Caller** — `GET /workspaces/{id}/views/{vendors_view}/data` with criteria `"Vendor Name"='{name}'`
- **Invoice Status** — `GET /workspaces/{id}/views/{invoices_view}/data` with criteria `"Load Number"='{number}' AND "Vendor Name"='{name}'`
- **Rate Verification** — `GET /workspaces/{id}/views/{rates_view}/data` with criteria `"Load Number"='{number}'`
- **Identify Location** — `GET /workspaces/{id}/views/{locations_view}/data` with criteria `"Company Name" LIKE '%{name}%'`

## 4.2 Architecture Diagrams

**Overall Sequence:**

```
sequenceDiagram
    participant V as Vendor (Caller)
    participant CA as ConvoAgent
    participant ZA as ZohoanalyticsAgent
    participant API as Zoho Analytics API

    Note over ZA: Setup (on publish)
    CA->>ZA: project_publish_finish
    ZA->>ZA: Create connectors, register tools
    ZA->>CA: Tools registered via update_newo_tools
```

```
    Note over V,API: Vendor calls about invoice
    V->>CA: "Hi, this is John from ABC Trucking..."
    CA->>CA: LLM detects verify_caller intent
    CA->>ZA: zohoanalytics_verify_caller_tool (memory)
    ZA->>ZA: LLM extracts vendor_name from memory
    ZA->>API: GET /views/{vendors}/data?CONFIG=...
    API-->>ZA: Vendor data (name, phone, contact)
    ZA->>CA: urgent_message: "Caller verified: ABC Trucking"
    CA->>V: "I've verified your identity..."

    V->>CA: "Check invoice for load LD-12345"
    CA->>CA: LLM detects invoice_status intent
    CA->>ZA: zohoanalytics_query_invoice_status_tool (memory)
    ZA->>ZA: LLM extracts load_number from memory
    ZA->>API: GET /views/{invoices}/data?CONFIG=...
    API-->>ZA: Invoice data (due date, status, amount, payee)
    ZA->>CA: urgent_message: "Invoice found: Pending, $3500, Apr 15"
    CA->>V: "Your invoice status is Pending, $3,500 due April 15th..."
```

**SetupFlow:**

```
flowchart TD
    E["project_publish_finish"] --> GUARD{"registry_item_idn ==
zohoanalytics_integration?"}
    GUARD -->|"No"| RET["Return()"]
    GUARD -->|"Yes"| REGION["Resolve regional URLs from zohoanalytics_region"]
    REGION --> INIT["_init: register attributes, create connectors, create persona"]
    INIT --> OAUTH{"OAuth code provided?"}
    OAUTH -->|"Yes"| TOKENS["_getTokensSkill: exchange code for tokens"]
    OAUTH -->|"No"| REFRESH{"Refresh token exists?"}
    REFRESH -->|"Yes"| RTOKEN["_refreshTokenSkill: refresh access token"]
    REFRESH -->|"No"| TOOLS["_setupToolsSkill: register custom tools"]
    TOKENS --> TOOLS
    RTOKEN --> TOOLS
    TOOLS --> DONE["Setup complete"]
```

**CallerVerificationFlow:**

```
flowchart TD
    E["zohoanalytics_verify_caller_tool"] --> EXTRACT["_extractCallerInfoSkill: LLM
extracts vendor_name, phone from memory"]
    EXTRACT --> BUILD["_verifyCallerSkill: build Sync Export query"]
    BUILD --> UTILS["zohoanalytics_execute_request → UtilsFlow"]
    UTILS --> HTTP["GET /views/{vendors_view}/data?CONFIG=..."]
    HTTP --> RESULT{"HTTP 200?"}
    RESULT -->|"Error"| ERR["urgent_message: technical issue, offer transfer"]
    RESULT -->|"Success"| PARSE["_verifyCallerSuccessSkill: parse response"]
    PARSE --> FOUND{"Vendor found?"}
    FOUND -->|"Yes"| VERIFIED["Set caller_verified=true, urgent_message: Caller
verified"]
```

```
    FOUND -->|"No"| NOTVERIFIED["Set caller_verified=false, urgent_message:
Verification failed"]
```

**DataQueryFlow (Invoice Status):**

```
flowchart TD
    E["zohoanalytics_query_invoice_status_tool"] -->
EXTRACT["_extractQueryInfoSkill: LLM extracts load_number from memory"]
    EXTRACT --> VENDOR["Override vendor_name from caller_verified session"]
    VENDOR --> BUILD["_queryInvoiceStatusSkill: build Sync Export query"]
    BUILD --> UTILS["zohoanalytics_execute_request → UtilsFlow"]
    UTILS --> HTTP["GET /views/{invoices_view}/data?CONFIG=..."]
    HTTP --> RESULT{"HTTP 200?"}
    RESULT -->|"Error"| ERR["urgent_message: technical issue"]
    RESULT -->|"Success"| PARSE["_queryInvoiceStatusSuccessSkill: parse response"]
    PARSE --> FOUND{"Invoice found?"}
    FOUND -->|"Yes"| DATA["urgent_message: due date, status, amount, payee"]
    FOUND -->|"No"| NOTFOUND["urgent_message: no invoice found"]
```

**DataQueryFlow (Rate Verification):**

```
flowchart TD
    E["zohoanalytics_query_rate_verification_tool"] -->
EXTRACT["_extractQueryInfoSkill: LLM extracts load_number from memory"]
    EXTRACT --> BUILD["_queryRateVerificationSkill: build Sync Export query"]
    BUILD --> UTILS["zohoanalytics_execute_request → UtilsFlow"]
    UTILS --> HTTP["GET /views/{rates_view}/data?CONFIG=..."]
    HTTP --> RESULT{"HTTP 200?"}
    RESULT -->|"Error"| ERR["urgent_message: technical issue"]
    RESULT -->|"Success"| PARSE["_queryRateVerificationSuccessSkill: parse
response"]
    PARSE --> FOUND{"Rate found?"}
    FOUND -->|"Yes"| DATA["urgent_message: rate, payee, payment amount"]
    FOUND -->|"No"| NOTFOUND["urgent_message: no rate found"]
```

**UtilsFlow:**

```
flowchart TD
    E["zohoanalytics_execute_request"] --> EXEC["_executeRequestSkill: inject Zoho-
oauthtoken + ZANALYTICS-ORGID headers"]
    EXEC --> HTTP["SendCommand to zohoanalytics_connector"]
    HTTP --> RESP{"HTTP Response"}
    RESP -->|"401"| REFRESH["_refreshTokenSkill: POST to token endpoint"]
    REFRESH --> SAVE["_saveTokenSkill: store new access_token"]
    SAVE --> RETRY{"Attempt < 3?"}
    RETRY -->|"Yes"| EXEC
    RETRY -->|"No"| FAIL["zohoanalytics_result_error"]
    RESP -->|"Body error"| FAIL
    RESP -->|"Success"| OK["zohoanalytics_result_success"]
```

### 4.3 Where to Test

Testing can be done through the **Newo conversation interface** (chat or voice channel) by interacting with the agent as a vendor calling about invoices or rates.

### 4.4 Testing and Verification

**To test the integration:**

1. **Setup**: Publish the project and verify attributes appear in the builder (Client ID, Region, View IDs)
2. **OAuth**: Click the authorization link in OAuth Code attribute, authorize, paste code, publish again. Verify `zohoanalytics_access_token` is populated (hidden attribute)
3. **Caller Verification**: Say "Hi, this is [vendor name from Vendors table]" → Agent should say "Let me verify your information" → Confirm verification succeeded
4. **Invoice Lookup**: Ask "What's the status of my invoice for load [load number]?" → Agent should return due date, payment status, amount, payee
5. **Rate Verification**: Ask "What's the rate for load [load number]?" → Agent should return rate, payee, payment amount

**If no action occurs:**

- Ensure Client ID and Client Secret are set before first Publish
- Verify Region matches your Zoho account data center (check URL: `.zoho.com` = US, `.zoho.eu` = EU, etc.)
- Confirm all View IDs are filled (check for empty values)
- Re-publish after any attribute changes
- Check that table column names match exactly: `Load Number`, `Vendor Name`, `Due Date`, `Payment Status`, `Amount Due`, `Payee`, `Rate`, `Payment Amount`, `Phone`, `Contact Person`, `Email`, `Company Name`, `Location`, `Region`, `Account ID`
- Verify data exists in Zoho Analytics tables (empty tables return "not found")

> *Note: This integration performs **read-only** queries against Zoho Analytics. No data is modified in Zoho Analytics through the agent.*

## 5. FAQ

**Q: Does the integration modify data in Zoho Analytics?** A: No. The integration is read-only — it only queries data via the Sync Export API. No rows are created, updated, or deleted.

**Q: How does data get into Zoho Analytics tables?** A: Data must be synced from your TMS (Transportation Management System) separately. Zoho Analytics supports imports from databases, CSV, cloud storage, and 250+ integrations. Data freshness depends on your sync frequency.

**Q: What happens if the vendor is not found during verification?** A: The agent informs the caller that verification failed and offers to transfer to a human representative.

**Q: Can the agent handle multiple invoice lookups in one call?** A: Yes. After answering an invoice question, the agent asks "Is there anything else?" The caller can ask about another load number.

**Q: What if the table has more than 1 million rows?** A: The Sync Export API only works for tables under 1 million rows. For larger tables, an async export approach would be needed (not currently implemented).

**Q: Which Zoho regions are supported?** A: US, EU, India, Australia, Japan, Saudi Arabia, and Canada. Set the correct region in the `zohoanalytics_region` attribute.

**Q: Can I use Zoho Analytics free tier?** A: Yes. The free tier (2 users, 5 workspaces, 10,000 rows) is sufficient for testing and small deployments.